

Education

PhD. Biostatistics, University of Washington, Seattle

– *Expected 2026*

MSc. Statistics and Data Science, University of Central Florida, Orlando

BSc. Mathematics, Tufts University, Medford

Statistical Expertise

Causal inference, Non/Semi-parametric inference, Statistical/Machine learning, Deep learning, Treatment effect heterogeneity, High dimensional methods, Survival analysis.

Research Experience

University of Washington, Seattle

September 2023 – present

Research Assistant, Advisors: Prof. Ali Shojaie & Dr. Aaron Hudson

- Developed a general non-parametric framework for testing hypotheses about a class of function-valued parameters. Special application; inference on the conditional average treatment effect (CATE).
- Evaluating conditional distributional treatment effect using Generative adversarial network (GAN).

Fred Hutchinson Cancer Center, Seattle

February 2023 – present

Research Assistant, Advisor: Prof. Ying-Qi Zhao

- Developed optimal biomarker combination decision rules that maximizes patient benefit under a pre-specified positive predictive value. Particular applications in rare event detection (e.g., pancreatic cancer), surveillance and diagnostic.
- External information transfer; incorporated external risk information to improve new decision rule.
- Testing cancer screening trial efficacy using change in late-stage events.
- Methods for integrating real world data with experimental data to estimate treatment effect.

Data Science Lab, University of Central Florida, Orlando

January 2021 – May 2021

Deep One-Class Classification | Anomaly Detection

- Proposed an approach to anomaly detection that incorporates deep networks in minimizing the objective function.
- Implemented the method in R software to solve the formulated constrained optimization problems with gradient descent.

Biostatistics Lab, University of Central Florida, Orlando

May 2020 – May 2021

Biostatistician, Advisor: Prof. Julia Soulakova

- We studied the effect of mild Hypobaric Hypoxia exposure on Splenic proliferating, Erythroid, and Myeloid cells after muscle Trauma-Hemorrhage. Research funded by the Department of Defense.
- Conducted statistical analyses on data from phase I trial, presented analytical results to non-statistical collaborators, and proposed statistical methods. Utilized SAS[®] software to write programs for analyses and data management.

Publications

- **Osom, Albert** & Zhao, Ying-Qi. Developing Performance-Guaranteed Biomarker Combination Rules with Integrated External Information under Practical Constraints (*2025 ASA Risk Analysis Section Student Paper Award*).
- **Osom, Albert** & Hudson, Aaron & Shojaie, Ali. A general non-parametric framework for testing hypotheses about a class of function-valued parameters (*to be submitted 2025+*).
- **Osom, Albert** & Barton, Krysta & Sexton, Katie & Brumback, Lyndia & Yi-Frazier, Joyce & Rosenberg, Abby & Engelberg, Ruth & Steiner, Jill. (2024). Impact of Social Determinants of Health on Self-Perceived Resilience: An Exploratory Study of Two Cohorts of Adults with Congenital Heart Disease. *Congenital Heart Disease* 19.1 (2024): 33-48.
- Steiner, J. M., **Osom, A.**, Brumback, L., Barton, K., Sexton, K., Rosenberg, A., & Curtis, J. R. (2023). IMPACT OF SOCIAL DETERMINANTS OF HEALTH ON RESILIENCE IN ADULTS WITH CONGENITAL HEART DISEASE. *Journal of the American College of Cardiology*, 81(8-Supplement), 1582-1582.
- Liyuan Zhang, Shailey Patel, **Albert Osom**, Julia Soulakova, Charles C. Caldwell, Barbara St. Pierre Schneider. The Effect of Mild Hypobaric Hypoxia on Splenic Proliferating, Erythroid, and Myeloid Cells After Muscle Trauma-Hemorrhage. {*Submitted*}

Presentations

- 2020 Nursing Summit: Modified Interquartile Range method for Detecting Extreme Outliers in Small Samples. August, 28th. {*Invited Talk*}.
- 2021 UCF Student Scholar Symposium: Four Methods for Detecting Outliers: Applications in Animal Study. Osom, A. Zhang, L., St. Pierre Schneider and Soulakova, J.N., Orlando, FL, March 30, 2021.

Computational/Technical Skills

R, Python, SAS[®], SPSS, SQL, L^AT_EX, Spark